Indian Institute of Technology, Kanpur

ECO702: Macroeconomics (3-0-0-0 (9))

Neelanjan Datta

Course Handout

1. **Contact Information:** ndatta@iitk.ac.in
2. **Office Hours:** Time slot to be decided; room 428 ESB-2 building.
3. **Lectures:** Mondays and Wednesdays, 2 PM – 3:15 PM, L02
4. **Pre-requisites:** None

5. **Course Description:**

This Ph.D. level course in macroeconomics is designed to provide an in-depth exploration of advanced topics and contemporary research in the field. Aimed at doctoral/advanced masters students with an existing foundation in macroeconomic theory, this course emphasizes critical thinking, and development of analytical skills. The course is designed to include microeconomic foundations of existing macroeconomic models.

a. **Objectives:**

Upon completion of this course, students should be able to:

1. Demonstrate a deep understanding of the theoretical foundations of macroeconomics.
2. Develop the ability to read, understand, and critique peer-reviewed research articles in the field, and apply this knowledge to formulate original research.

b. **Course Content:**

1. **Representative Agent Model –** Static and dynamic versions, elements of macroeconomic models-preferences, endowment, technology, optimization, competitive equilibrium, pareto optima, welfare theorems
2. **Dynamic Optimization –** Inter-temporal preferences, basics of dynamic optimization, consumption smoothing and comparative statics, envelope theorem, contraction mapping theorem
3. **Overlapping Generations Model –** Life-cycle consumers and demographics, equilibrium, steady-state and dynamics, borrowing restrictions
4. **Long-run Growth in the Economy –** Growth accounting, Solow-growth model, technological progress in the Solow growth model, one-sector growth model, endogenous growth theory, stochastic growth
5. **Real Business-Cycle Model –** Rational expectations hypothesis, New Keynesian Dynamic Stochastic General Equilibrium model, monetary policy, inflation, criticisms and new developments

6. **Grading Policy:**

The total grade from the course will be distributed among three components. The points will be allocated as follows:

a. **Mid-semester Exam (30%) –** This will cover topics taught till the week before the mid-semester exams week.
b. **Assignment (20%) –** More details to follow; this will be due at the beginning of April (specific date will be announced during lecture).
c. **Final Exam (50%) -** Comprehensive exam covering all topics taught in the course; to be held during the final exams week

7. **Suggested text and reference material:**

Detailed lecture notes will be provided. The following books are great references. Book 3 is the main reference for modules 1-4. Book 1 is the main reference for module 5. Parts of module 2 will be drawn from book 4.

1. Gali, Jordi, *Monetary Policy, Inflation, and the Business Cycle*, Princeton University Press
2. Romer, David, *Advanced Macroeconomics,* McGraw-Hill
3. Ljungqvist, L., and T. J. Sargent, *Recursive Macroeconomic Theory,* MIT Press
4. Stokey, N., Lucas Jr., R, and E. Prescott, *Recursive Methods in Economic Dynamics,* Harvard University Press
5. Barro, R. J., and X. Sala-i-Martin, *Economic Growth*, MIT Press
6. Heijdra, J Ben, *The Foundations of Modern Macroeconomics,* Oxford University Press
7. Wickens, Michael, *Macroeconomic Theory,* Princeton University Press
8. Williamson, Stephen D., *Macroeconomics*, Pearson
9. Acemoglu, D., Laibson, D., and J.A. List, *Macroeconomics*, Pearson

8. Additional Information:

a. **Copyright:** The instructor of this course owns the copyright of all the lecture materials. The lecture materials are distributed only to the students attending the course (ECO702) at IIT Kanpur and should not be distributed in print or through electronic media without the consent of the instructor. Students can make their own copies of the course material for their use.

b. **Missing Deadlines:** Students need to have a valid documented excuse and notify me in advance if you are missing a specific deadline. The validity of the excuse is entirely within my discretion.

c. **Academic Misconduct:** This includes all forms of student academic misconduct wherever committed; illustrated by, but not limited to, cases of plagiarism and dishonest practices in connection with examinations. Any form of academic misconduct will lead to an automatic F in the course.